

Raghav Gaba

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SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++

Frontend: Next.js, React.js, Tailwind CSS

Backend & Databases: Node.js, Express.js, PostgreSQL, MongoDB, Redis, Firebase, Appwrite, WebSockets

Tools & Platforms: Git, GitHub, Docker, Postman, OpenRouter

PROJECTS

Inlaya - AI-Assisted Software Engineering Platform

(inlaya.raghavgaba.me |

github.com/raghavvgaba/inlaya)

Tech Stack: Next.js, React, TypeScript, PostgreSQL, Prisma, Docker, E2B, GitHub API

- Engineered an **end-to-end AI-assisted development platform** that transforms GitHub issues into isolated coding workspaces with persistent conversations, live application previews, repository editing, and pull-request submission.
- Designed a **custom LLM agent runtime** with six repository tools, typed function calling, bounded file operations, structured responses, failure recovery, and enforced read-only **Plan** and write-enabled **Build** modes.
- Built a **recoverable E2B sandbox lifecycle** that clones repositories, detects project configurations, installs dependencies, launches application previews, persists session metadata, and reconnects active development environments after application reloads.
- Automated the **GitHub issue-to-pull-request workflow** using GitHub OAuth and GitHub Apps, including change detection, deterministic branch creation, authenticated commits, remote pushes, and pull-request creation or reuse.

Mosaic - Note Taking App

(mosaic.raghavgaba.me | github.com/raghavvgaba/mosaic)

Tech Stack: Next.js 16, TypeScript, Tailwind CSS, OpenRouter API, Zustand, SWR, Appwrite, Qdrant

- Architected and shipped a **production-grade AI-powered note-taking platform** with secure authentication and persistent document storage using Appwrite.
- Integrated **streaming LLM workflows via OpenRouter API**, enabling multi-model draft generation and summarization with real-time token streaming.
- Built **semantic search using 1536-dimensional OpenAI embeddings with Qdrant (cosine similarity)**, enabling scalable vector indexing and context-aware note retrieval.
- Engineered a **Notion-style block editor** with drag-and-drop architecture, slash-command parsing, and full document/workspace CRUD flows.
- Optimized client-server state with **SWR**, implementing caching, background revalidation, and optimistic updates for low-latency UX.

Chess — Real-Time Multiplayer Platform (chess.raghavgaba.me | github.com/raghavvgaba/chess)

Tech Stack: React, TypeScript, Node.js, Express, WebSockets, Redis, PostgreSQL, Better Auth, chess.js

- Engineered a **production-ready real-time chess platform** using a server-authoritative architecture, ensuring validated moves and consistent game state across live multiplayer sessions.
- Architected a **resilient state management system** using Redis for in-memory gameplay and PostgreSQL for durability, with an async flusher and retry-backoff pipeline.
- Designed a **robust matchmaking system** with queue orchestration, private room codes, TTL-based lifecycle management, and race-condition-safe validation.
- Developed a **full-featured user experience** with authentication, REST APIs, analytics dashboard, social/friend systems, and Stockfish bot integration.

EDUCATION

Manipal University Jaipur

B.Tech in Computer Science and Engineering

Jaipur, Rajasthan

2023 – 2027